

Laurens Bolle

SOFTWARE ARCHITECT & AI ENGINEER



SUMMARY | |

Laurens studied Informatics at the University of Hasselt and graduated with distinction from the University of Leuven in the Master of Computer Science. This education provided him with a broad theoretical and technical knowledge to function as an engineer in any environment.

Over the past eleven years, Laurens has worked on diverse projects, constantly encountering and mastering new technologies. His experience spans on-premise, cloud, and hybrid environments, giving him deep insights into the benefits and challenges of each approach. These experiences have made him a strong advocate for cloud-first development.

The past four years, AI has thoroughly changed the world of software development. Laurens was there from the beginning and immediately included it in his development workflows. As AI progressed and evolved, so did his way of working. Today, running fully agentic software development loops are no exception.

In his free time, Laurens enjoys tinkering with his 3D printer. He loves solving small household problems by designing and printing custom solutions. Helping friends and family by giving them a small 3D printed gift, and being able to show and tell them about the technology is even better!

KEY PROJECTS | |

Laurens is currently working as a full stack developer and architect at Waterleau, designing and implementing modern cloud-first application. He coaches and mentors his team on using AI effectively.

At Atrias, Laurens lead the development of a certification and load-testing tool for the Belgian energy market. Integration with legacy software and building a high-throughput messaging system in the cloud proved to be the key challenges.

KEY SKILLS | |

- .NET, C#
- Unit & integration testing, TDD
- OOP, DDD & SOLID principles
- Angular, Typescript
- Automating build & release pipelines
- Infrastructure as Code
- Ability to develop and maintain qualitative, unit tested, front- and back-end code using AI
- Coaching other developers through code reviews and pair programming
- Motivating teams and moving them into a continuous improvement mindset

- Bachelor of Informatics, Universiteit Hasselt, 09/2010 – 07/2013
- Master of Engineering: Computer Science, University of Leuven, 09/2013 – 01/2016

WORK EXPERIENCE | |

DEVELOPMENT LEAD, ARCHITECT & AI EVANGELIST

WATERLEAU : MAY 2022 – NOW

Rebuilding the proof-of-concept of 'Smartlab' into a production-worthy application. Smartlab is a cloud-based digital platform that transforms water and wastewater treatment facilities into IoT-enabled hubs. By integrating with existing PLC and SCADA systems, it provides real-time data access, advanced analytics, and AI-driven tools to optimize operational efficiency, reduce downtime, and ensure compliance with Environmental, Social, and Governance (ESG) standards.

Challenges:

- Creating an alternative data flow to redirect data from legacy plants into the new architecture, while staying operational
- Working with OT to standardize data structures to enable the creation of flexible and reusable dashboards
- Providing value to customers with limited budget and resources, by focusing on the minimum viable product
- Feeding data into a Digital Twin for real-time and scenario-based simulations

Tools & technologies: .NET 8 (C#), Angular 16 (Typescript), PostgreSQL, Entity Framework, Docker, Microsoft Azure (App Service, Container Apps, Event Hubs, Event Grids, Functions, Service Bus, Logic Apps), Terraform, Azure Databricks, Inductive Automation Ignition, Python, Azure DevOps, JIRA, REST APIs with OpenAPI specifications

FULL STACK DEVELOPER & SCRUM MASTER

FARMAD : AUGUST 2021 – APRIL 2022

Designing and implementing a modern, cloud-first replacement of the legacy 'Twin' application. Farmad provides software for pharmacists to enable their day-to-day operations.

Challenges:

- Migrating data from the legacy software to the new cloud platform, and coping with its bad data quality
- Building a true MVP and avoiding scope creep, as time-to-market is key for the success of the project
- Introducing a DDD approach and coaching new developers who are not familiar with its principles

Tools & technologies: .NET, Angular, PostgreSQL, Docker, AWS (S3, EC2 container instances), Terraform, Bitbucket, JIRA

LEAD DEVELOPER

ATRIAS : NOVEMBER 2020 – JULY 2021

Implementation of a custom test tool on top of Azure PaaS/FaaS to easily define and execute both functional acceptance tests and high-load NFR tests of the Central Market System. Atrias acts as the central hub in the Belgian energy market, supporting DGO's and suppliers. Since each actor in the energy flow could be defined as 'live' or 'stubbed' system by the test tool, the components could be reused by Atrias to support QA in other projects like supplier onboarding.

Job description:

- Design and implementation of application and cloud architecture until final delivery
- Support customer with deployment and cloud security aspects
- Coaching junior developers in the team

Tools & technologies: .NET 5, React, PostgreSQL, Docker, Azure, Terraform, Azure DevOps

FULL STACK DEVELOPER & SCRUM MASTER

SESVANDERHAVE : JULY 2018 – OCTOBER 2020

Development of a platform of separate microservices to enable the R&D department to "Scale-up & Accelerate the detection, selection and growing of desirable traits into new outperforming varieties of sugar beets".

Responsibilities included:

- Developing new features in a variety of business contexts (automated sowing robots, breeding, genetics, seed management)
- Coaching fellow developers to produce high-quality code
- Ensuring continuous improvement of the team through scrum rituals

Tools & technologies: .NET Core 3.1, Angular, Oracle